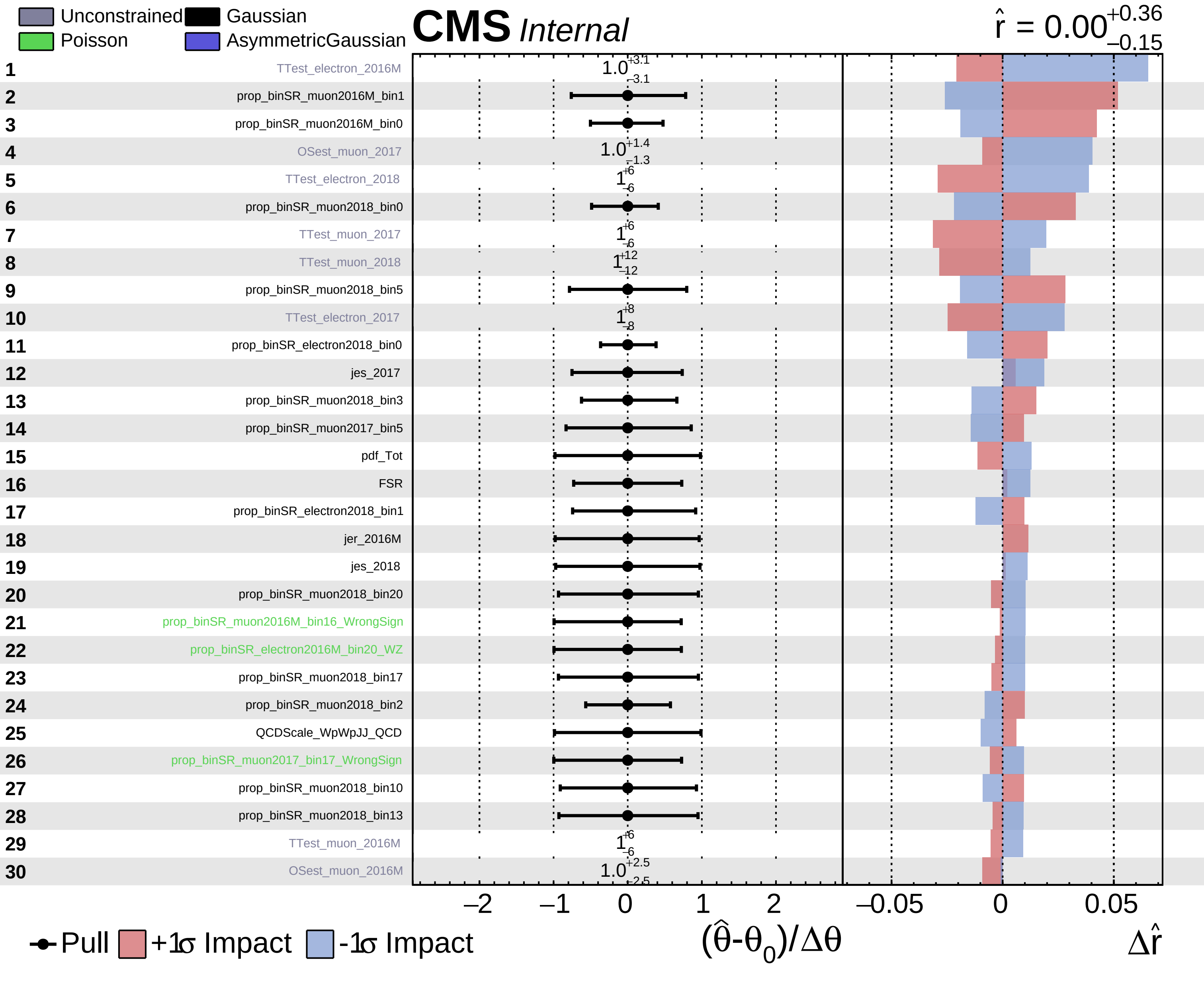
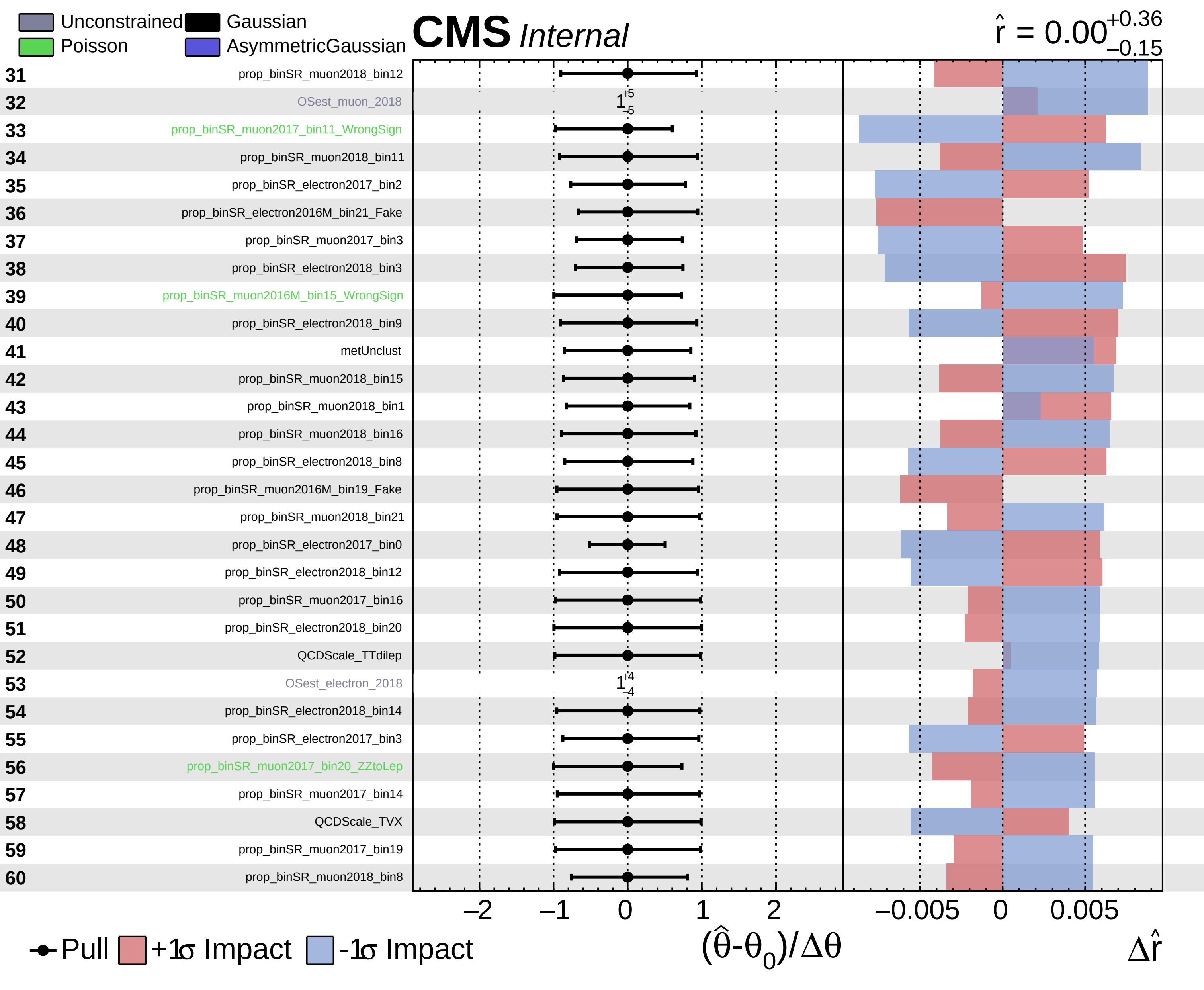
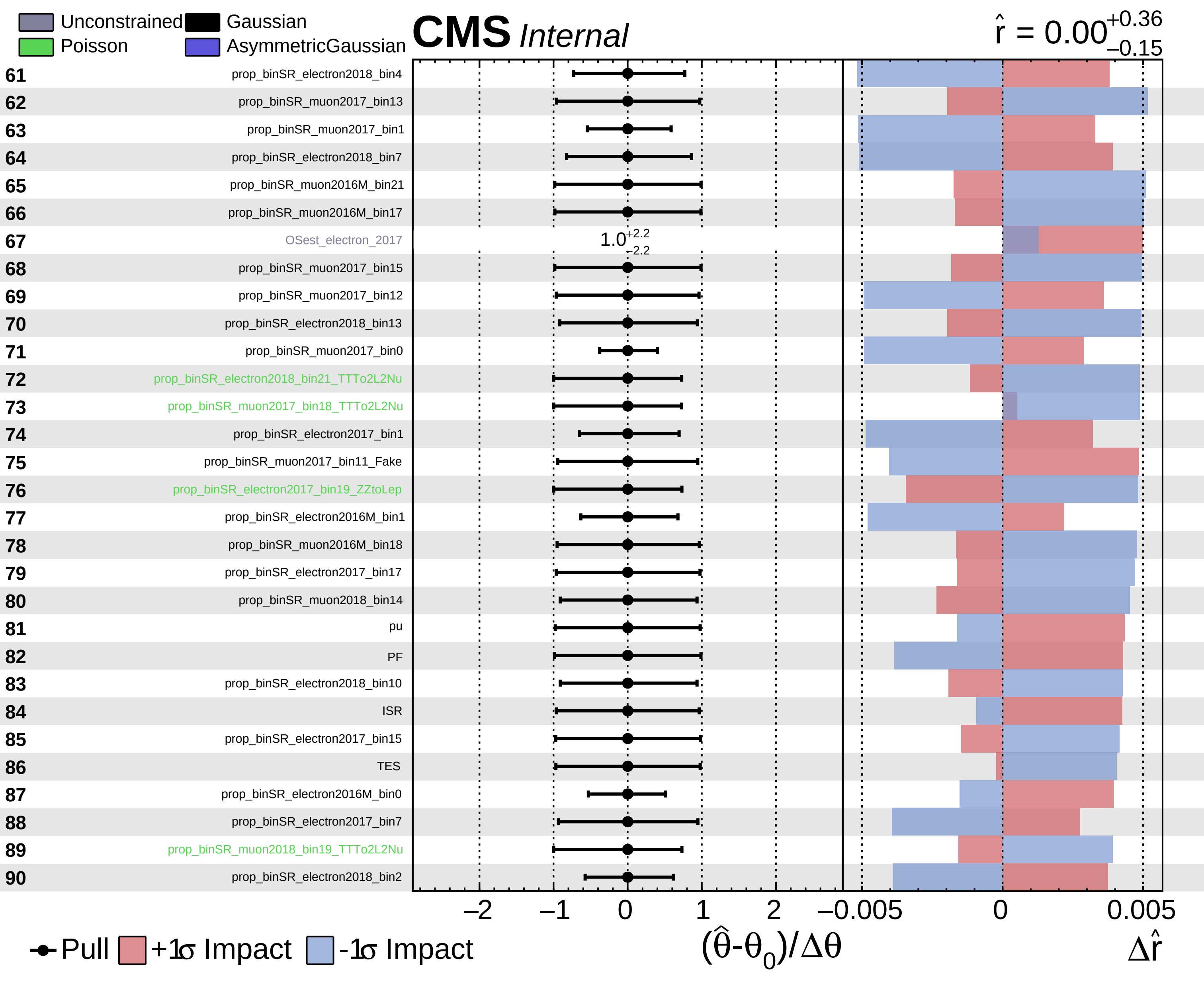
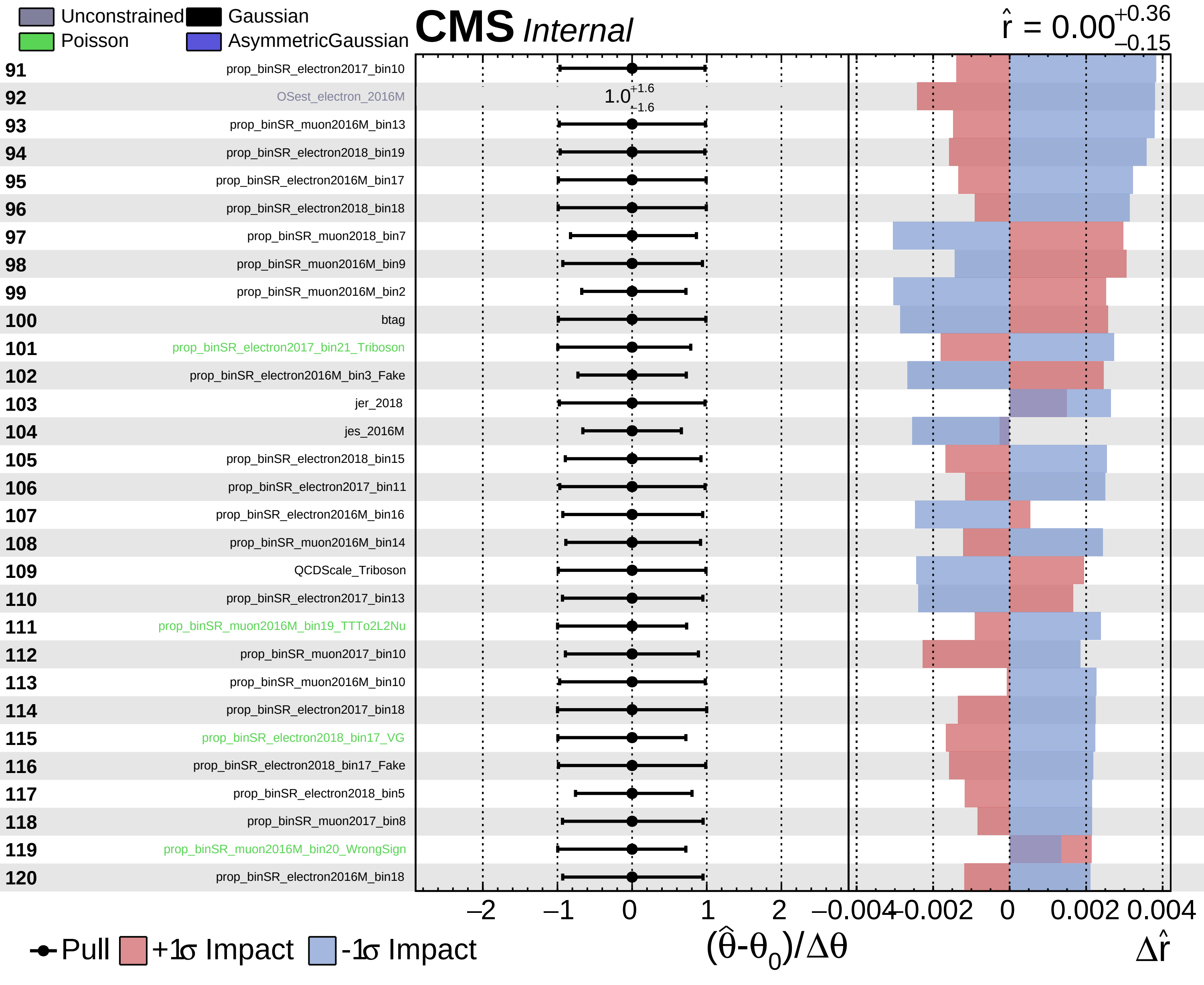
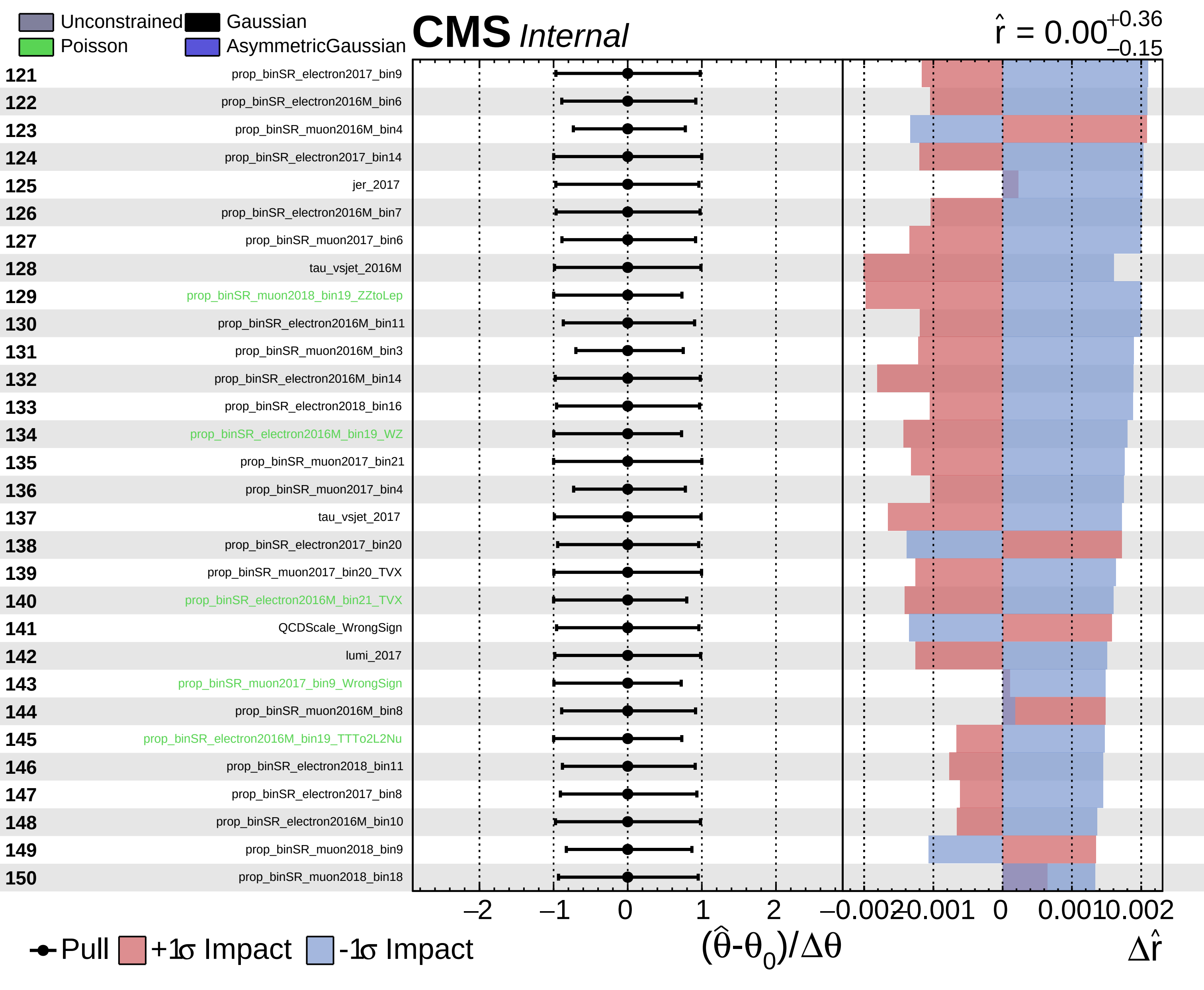


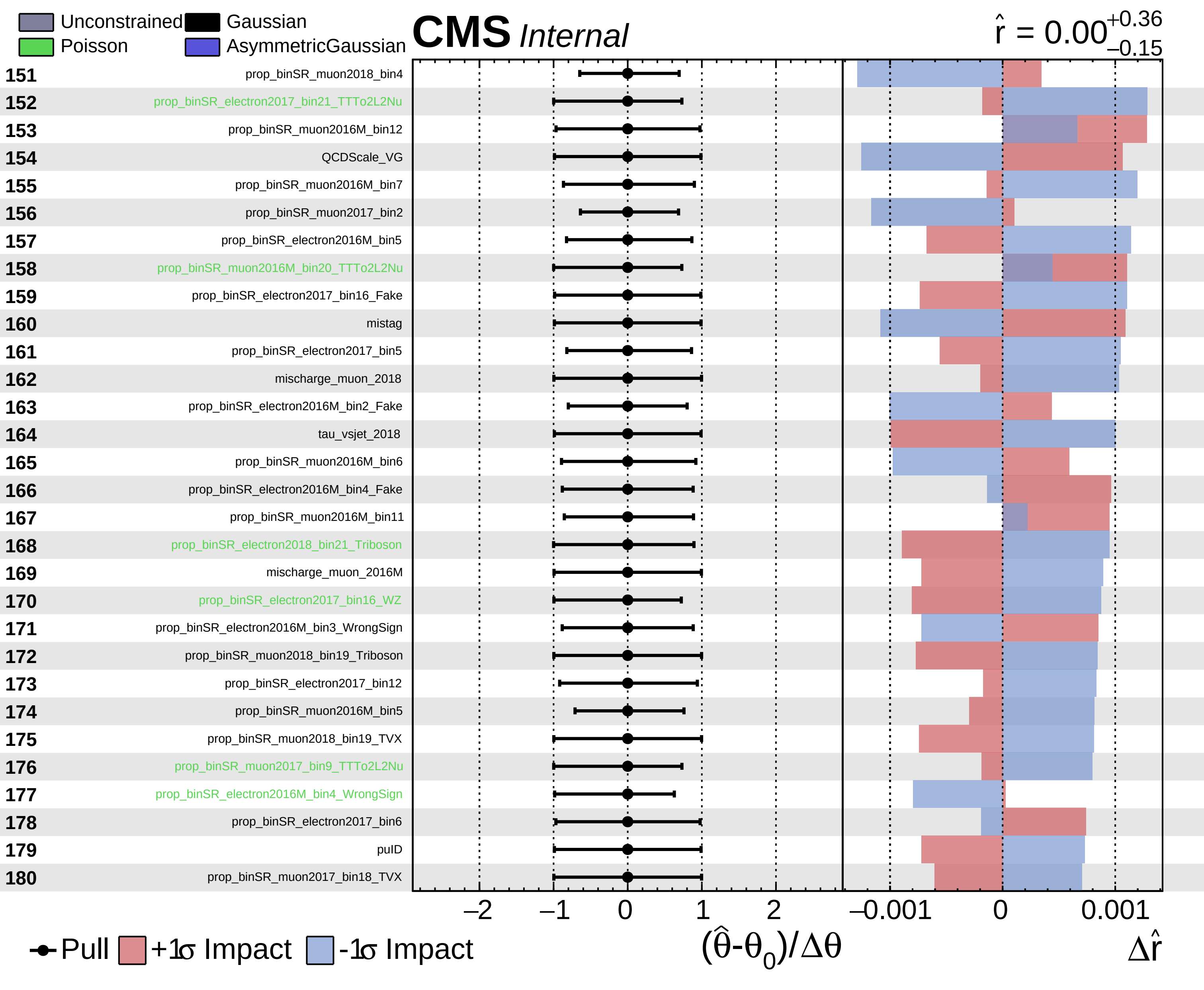
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.15
1 TTest_electron_2016M
2 prop_binSR_muon2016M_bin1
3 prop_binSR_muon2016M_bin0
4 OSest_muon_2017
5 TTest_electron_2018
6 prop_binSR_muon2018_bin0
7 TTest_muon_2017
8 TTest_muon_2018
9 prop_binSR_muon2018_bin5
10 TTest_electron_2017
11 prop_binSR_electron2018_bin0
12 jes_2017
13 prop_binSR_muon2018_bin3
14 prop_binSR_muon2017_bin5
15 pdf_Tot
16 FSR
17 prop_binSR_electron2018_bin1
18 jer_2016M
19 jes_2018
20 prop_binSR_muon2018_bin20
21 prop_binSR_muon2016M_bin16_WrongSign
22 prop_binSR_electron2016M_bin20_WZ
23 prop_binSR_muon2018_bin17
24 prop_binSR_muon2018_bin2
25 QCDScale_WpWpJJ_QCD
26 prop_binSR_muon2017_bin17_WrongSign
27 prop_binSR_muon2018_bin10
28 prop_binSR_muon2018_bin13
29 TTest_muon_2016M
30 OSest_muon_2016M
1.0 +3.1 −3.1
1.0 +1.4 −1.3
1 +6 −6
1 +6 −6
1 +12 −12
1 +8 −8
1 +6 −6
1.0 +2.5 −2.5
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.15
31 prop_binSR_muon2018_bin12
32 OSest_muon_2018
1 +5 −5
33 prop_binSR_muon2017_bin11_WrongSign
34 prop_binSR_muon2018_bin11
35 prop_binSR_electron2017_bin2
36 prop_binSR_electron2016M_bin21_Fake
37 prop_binSR_muon2017_bin3
38 prop_binSR_electron2018_bin3
39 prop_binSR_muon2016M_bin15_WrongSign
40 prop_binSR_electron2018_bin9
41 metUnclust
42 prop_binSR_muon2018_bin15
43 prop_binSR_muon2018_bin1
44 prop_binSR_muon2018_bin16
45 prop_binSR_electron2018_bin8
46 prop_binSR_muon2016M_bin19_Fake
47 prop_binSR_muon2018_bin21
48 prop_binSR_electron2017_bin0
49 prop_binSR_electron2018_bin12
50 prop_binSR_muon2017_bin16
51 prop_binSR_electron2018_bin20
52 QCDScale_TTdilep
53 OSest_electron_2018
1 +4 −4
54 prop_binSR_electron2018_bin14
55 prop_binSR_electron2017_bin3
56 prop_binSR_muon2017_bin20_ZZtoLep
57 prop_binSR_muon2017_bin14
58 QCDScale_TVX
59 prop_binSR_muon2017_bin19
60 prop_binSR_muon2018_bin8
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

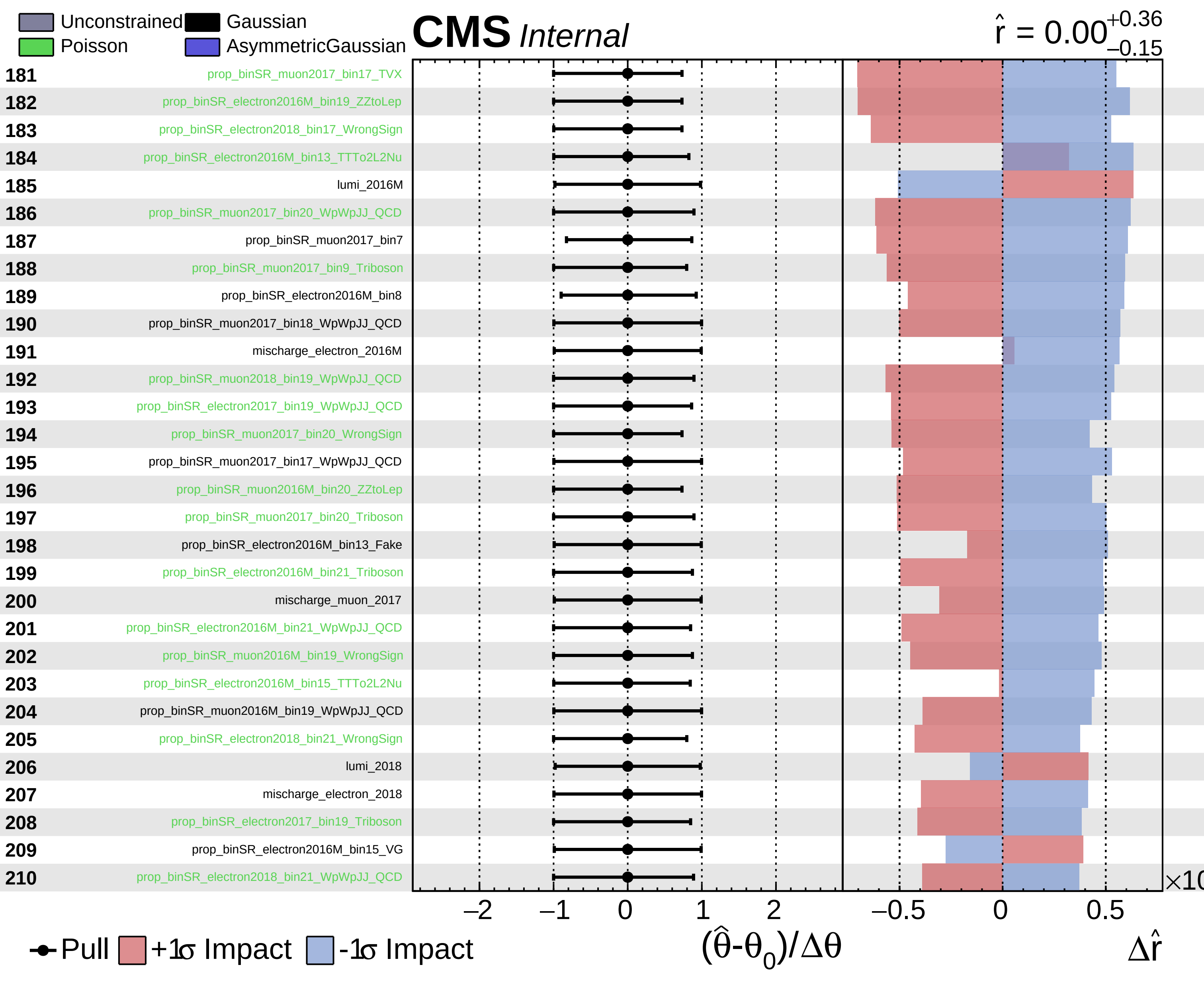

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
61 prop_binSR_electron2018_bin4
62 prop_binSR_muon2017_bin13
63 prop_binSR_muon2017_bin1
64 prop_binSR_electron2018_bin7
65 prop_binSR_muon2016M_bin21
66 prop_binSR_muon2016M_bin17
67 OSest_electron_2017
$1.0^{+2.2}_{-2.2}$
68 prop_binSR_muon2017_bin15
69 prop_binSR_muon2017_bin12
70 prop_binSR_electron2018_bin13
71 prop_binSR_muon2017_bin0
72 prop_binSR_electron2018_bin21_TTTo2L2Nu
73 prop_binSR_muon2017_bin18_TTTo2L2Nu
74 prop_binSR_electron2017_bin1
75 prop_binSR_muon2017_bin11_Fake
76 prop_binSR_electron2017_bin19_ZZtoLep
77 prop_binSR_electron2016M_bin1
78 prop_binSR_muon2016M_bin18
79 prop_binSR_electron2017_bin17
80 prop_binSR_muon2018_bin14
81 pu
82 PF
83 prop_binSR_electron2018_bin10
84 ISR
85 prop_binSR_electron2017_bin15
86 TES
87 prop_binSR_electron2016M_bin0
88 prop_binSR_electron2017_bin7
89 prop_binSR_muon2018_bin19_TTTo2L2Nu
90 prop_binSR_electron2018_bin2
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

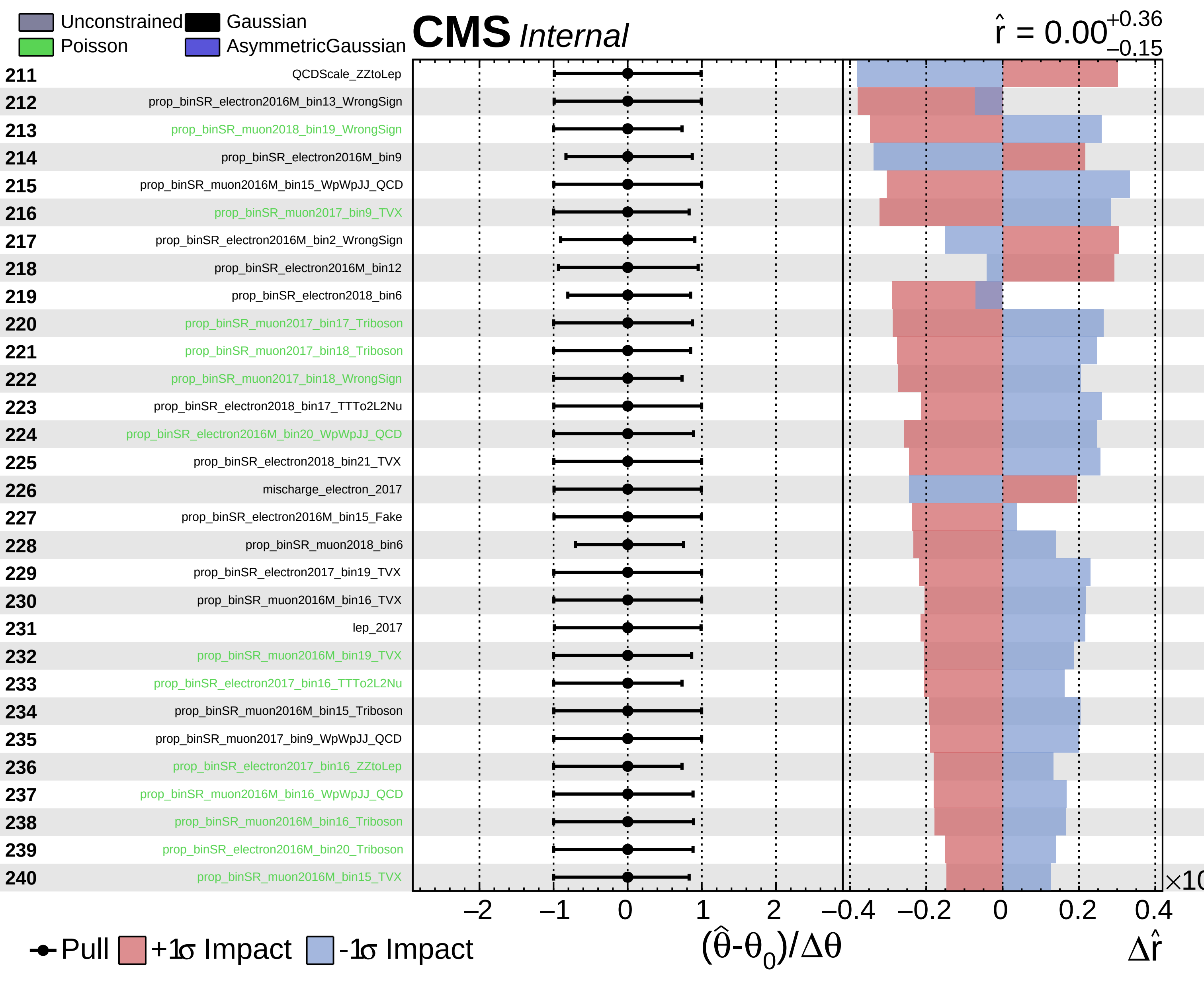

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
91 prop_binSR_electron2017_bin10
92 OSest_electron_2016M
$1.0^{+1.6}_{-1.6}$
93 prop_binSR_muon2016M_bin13
94 prop_binSR_electron2018_bin19
95 prop_binSR_electron2016M_bin17
96 prop_binSR_electron2018_bin18
97 prop_binSR_muon2018_bin7
98 prop_binSR_muon2016M_bin9
99 prop_binSR_muon2016M_bin2
100 btag
101 prop_binSR_electron2017_bin21_Triboson
102 prop_binSR_electron2016M_bin3_Fake
103 jer_2018
104 jes_2016M
105 prop_binSR_electron2018_bin15
106 prop_binSR_electron2017_bin11
107 prop_binSR_electron2016M_bin16
108 prop_binSR_muon2016M_bin14
109 QCDScale_Triboson
110 prop_binSR_electron2017_bin13
111 prop_binSR_muon2016M_bin19_TTTo2L2Nu
112 prop_binSR_muon2017_bin10
113 prop_binSR_muon2016M_bin10
114 prop_binSR_electron2017_bin18
115 prop_binSR_electron2018_bin17_VG
116 prop_binSR_electron2018_bin17_Fake
117 prop_binSR_electron2018_bin5
118 prop_binSR_muon2017_bin8
119 prop_binSR_muon2016M_bin20_WrongSign
120 prop_binSR_electron2016M_bin18
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
121 prop_binSR_electron2017_bin9
122 prop_binSR_electron2016M_bin6
123 prop_binSR_muon2016M_bin4
124 prop_binSR_electron2017_bin14
125 jer_2017
126 prop_binSR_electron2016M_bin7
127 prop_binSR_muon2017_bin6
128 tau_vsjet_2016M
129 prop_binSR_muon2018_bin19_ZZtoLep
130 prop_binSR_electron2016M_bin11
131 prop_binSR_muon2016M_bin3
132 prop_binSR_electron2016M_bin14
133 prop_binSR_electron2018_bin16
134 prop_binSR_electron2016M_bin19_WZ
135 prop_binSR_muon2017_bin21
136 prop_binSR_muon2017_bin4
137 tau_vsjet_2017
138 prop_binSR_electron2017_bin20
139 prop_binSR_muon2017_bin20_TVX
140 prop_binSR_electron2016M_bin21_TVX
141 QCDScale_WrongSign
142 lumi_2017
143 prop_binSR_muon2017_bin9_WrongSign
144 prop_binSR_muon2016M_bin8
145 prop_binSR_electron2016M_bin19_TTTo2L2Nu
146 prop_binSR_electron2018_bin11
147 prop_binSR_electron2017_bin8
148 prop_binSR_electron2016M_bin10
149 prop_binSR_muon2018_bin9
150 prop_binSR_muon2018_bin18
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

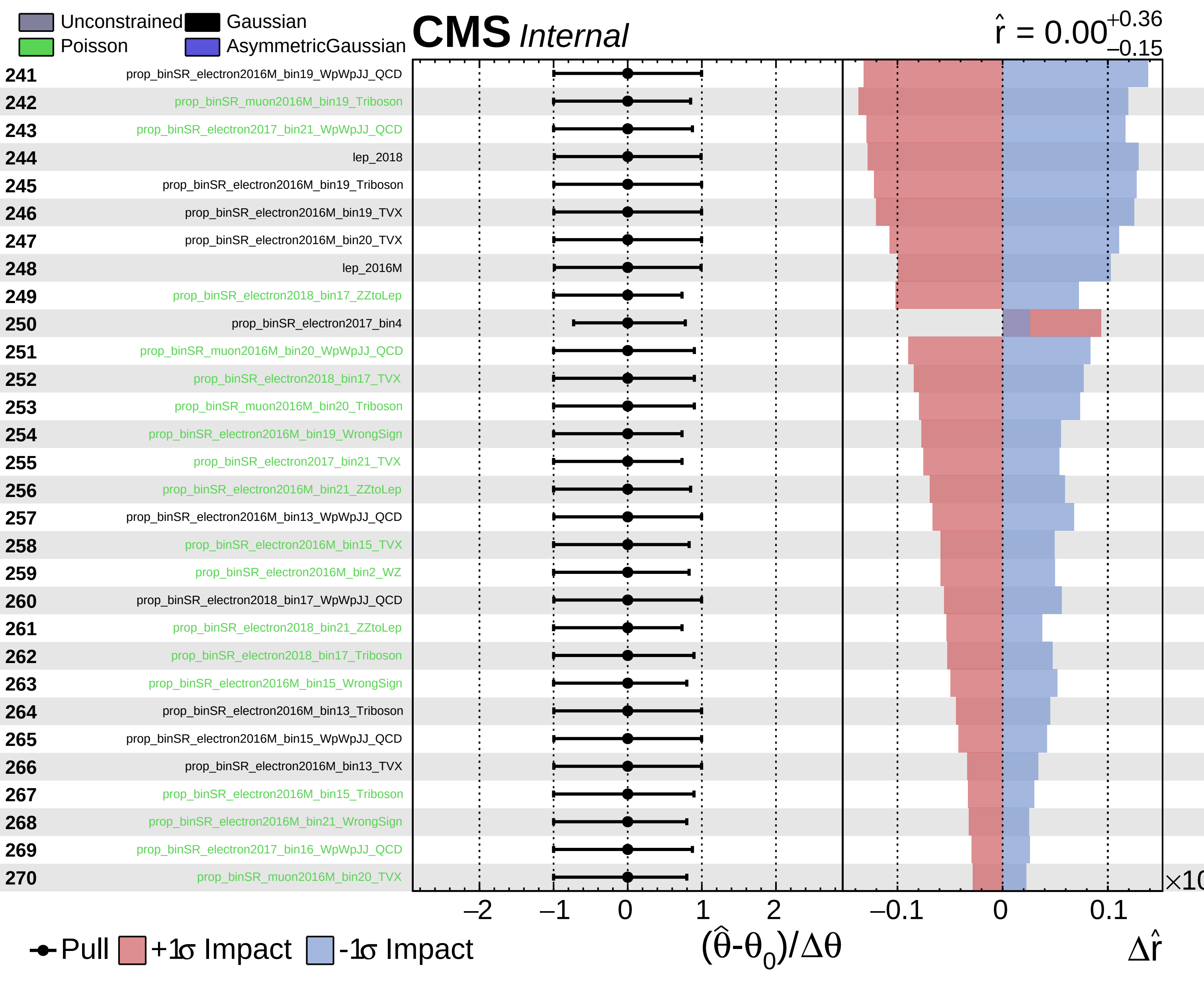


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
151 prop_binSR_muon2018_bin4
152 prop_binSR_electron2017_bin21_TTTo2L2Nu
153 prop_binSR_muon2016M_bin12
154 QCDScale_VG
155 prop_binSR_muon2016M_bin7
156 prop_binSR_muon2017_bin2
157 prop_binSR_electron2016M_bin5
158 prop_binSR_muon2016M_bin20_TTTo2L2Nu
159 prop_binSR_electron2017_bin16_Fake
160 mistag
161 prop_binSR_electron2017_bin5
162 mischarge_muon_2018
163 prop_binSR_electron2016M_bin2_Fake
164 tau_vsjet_2018
165 prop_binSR_muon2016M_bin6
166 prop_binSR_electron2016M_bin4_Fake
167 prop_binSR_muon2016M_bin11
168 prop_binSR_electron2018_bin21_Triboson
169 mischarge_muon_2016M
170 prop_binSR_electron2017_bin16_WZ
171 prop_binSR_electron2016M_bin3_WrongSign
172 prop_binSR_muon2018_bin19_Triboson
173 prop_binSR_electron2017_bin12
174 prop_binSR_muon2016M_bin5
175 prop_binSR_muon2018_bin19_TVX
176 prop_binSR_muon2017_bin9_TTTo2L2Nu
177 prop_binSR_electron2016M_bin4_WrongSign
178 prop_binSR_electron2017_bin6
179 puID
180 prop_binSR_muon2017_bin18_TVX
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

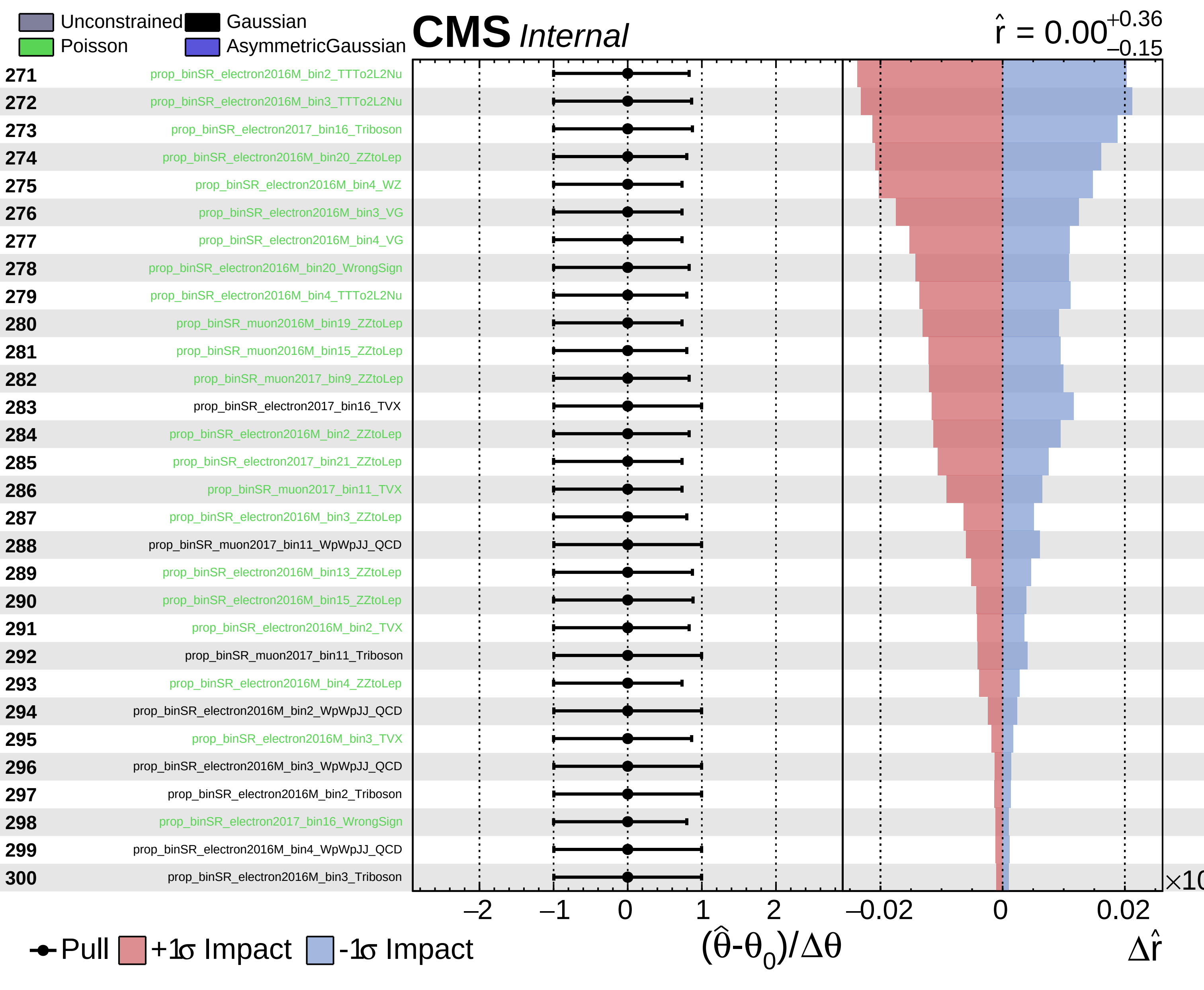

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
181 prop_binSR_muon2017_bin17_TVX
182 prop_binSR_electron2016M_bin19_ZZtoLep
183 prop_binSR_electron2018_bin17_WrongSign
184 prop_binSR_electron2016M_bin13_TTTo2L2Nu
185 lumi_2016M
186 prop_binSR_muon2017_bin20_WpWpJJ_QCD
187 prop_binSR_muon2017_bin7
188 prop_binSR_muon2017_bin9_Triboson
189 prop_binSR_electron2016M_bin8
190 prop_binSR_muon2017_bin18_WpWpJJ_QCD
191 mischarge_electron_2016M
192 prop_binSR_muon2018_bin19_WpWpJJ_QCD
193 prop_binSR_electron2017_bin19_WpWpJJ_QCD
194 prop_binSR_muon2017_bin20_WrongSign
195 prop_binSR_muon2017_bin17_WpWpJJ_QCD
196 prop_binSR_muon2016M_bin20_ZZtoLep
197 prop_binSR_muon2017_bin20_Triboson
198 prop_binSR_electron2016M_bin13_Fake
199 prop_binSR_electron2016M_bin21_Triboson
200 mischarge_muon_2017
201 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
202 prop_binSR_muon2016M_bin19_WrongSign
203 prop_binSR_electron2016M_bin15_TTTo2L2Nu
204 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
205 prop_binSR_electron2018_bin21_WrongSign
206 lumi_2018
207 mischarge_electron_2018
208 prop_binSR_electron2017_bin19_Triboson
209 prop_binSR_electron2016M_bin15_VG
210 prop_binSR_electron2018_bin21_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

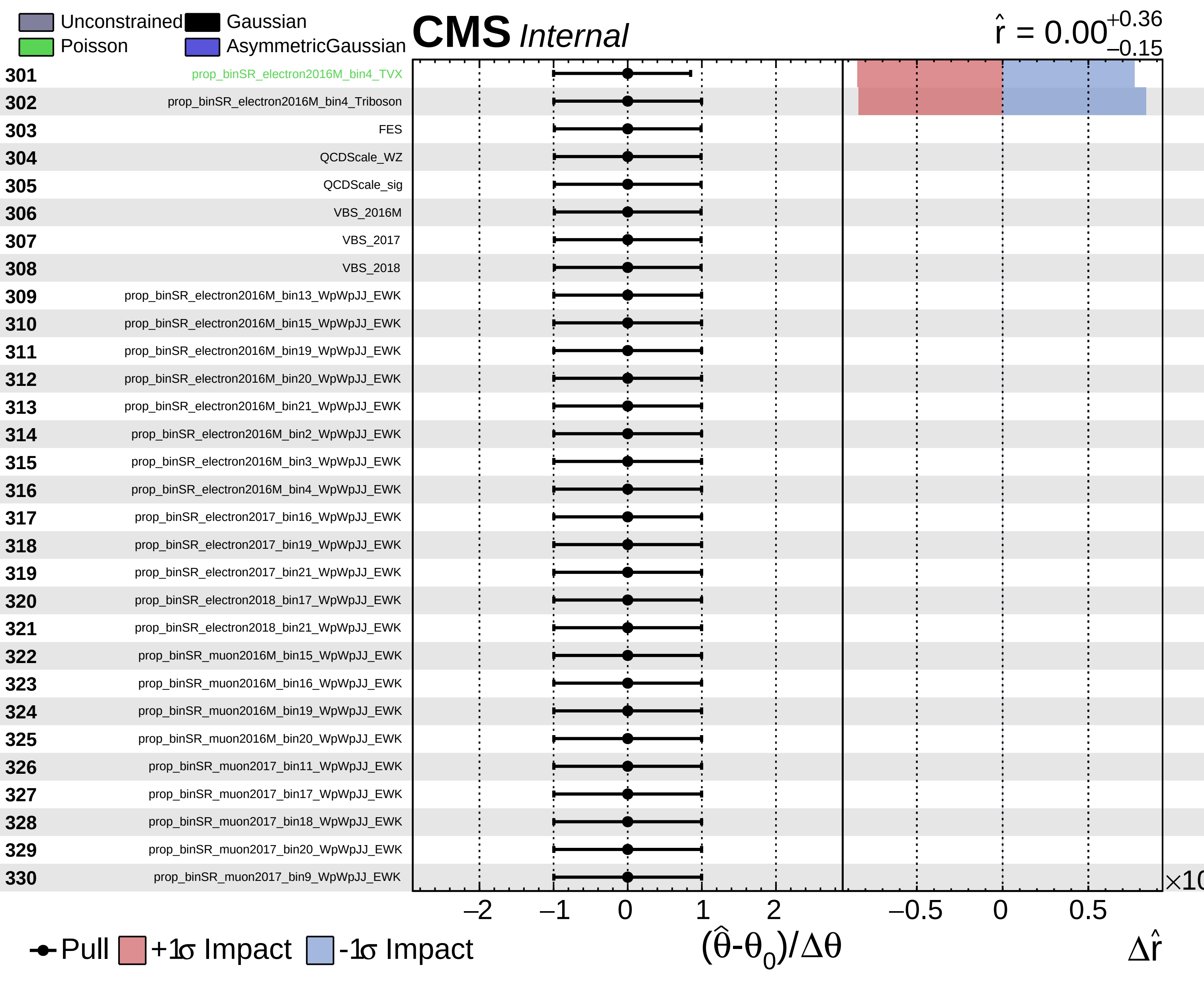


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
211 QCDScale_ZZtoLep
212 prop_binSR_electron2016M_bin13_WrongSign
213 prop_binSR_muon2018_bin19_WrongSign
214 prop_binSR_electron2016M_bin9
215 prop_binSR_muon2016M_bin15_WpWpJJ_QCD
216 prop_binSR_muon2017_bin9_TVX
217 prop_binSR_electron2016M_bin2_WrongSign
218 prop_binSR_electron2016M_bin12
219 prop_binSR_electron2018_bin6
220 prop_binSR_muon2017_bin17_Triboson
221 prop_binSR_muon2017_bin18_Triboson
222 prop_binSR_muon2017_bin18_WrongSign
223 prop_binSR_electron2018_bin17_TTTo2L2Nu
224 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
225 prop_binSR_electron2018_bin21_TVX
226 mischarge_electron_2017
227 prop_binSR_electron2016M_bin15_Fake
228 prop_binSR_muon2018_bin6
229 prop_binSR_electron2017_bin19_TVX
230 prop_binSR_muon2016M_bin16_TVX
231 lep_2017
232 prop_binSR_muon2016M_bin19_TVX
233 prop_binSR_electron2017_bin16_TTTo2L2Nu
234 prop_binSR_muon2016M_bin15_Triboson
235 prop_binSR_muon2017_bin9_WpWpJJ_QCD
236 prop_binSR_electron2017_bin16_ZZtoLep
237 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
238 prop_binSR_muon2016M_bin16_Triboson
239 prop_binSR_electron2016M_bin20_Triboson
240 prop_binSR_muon2016M_bin15_TVX
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

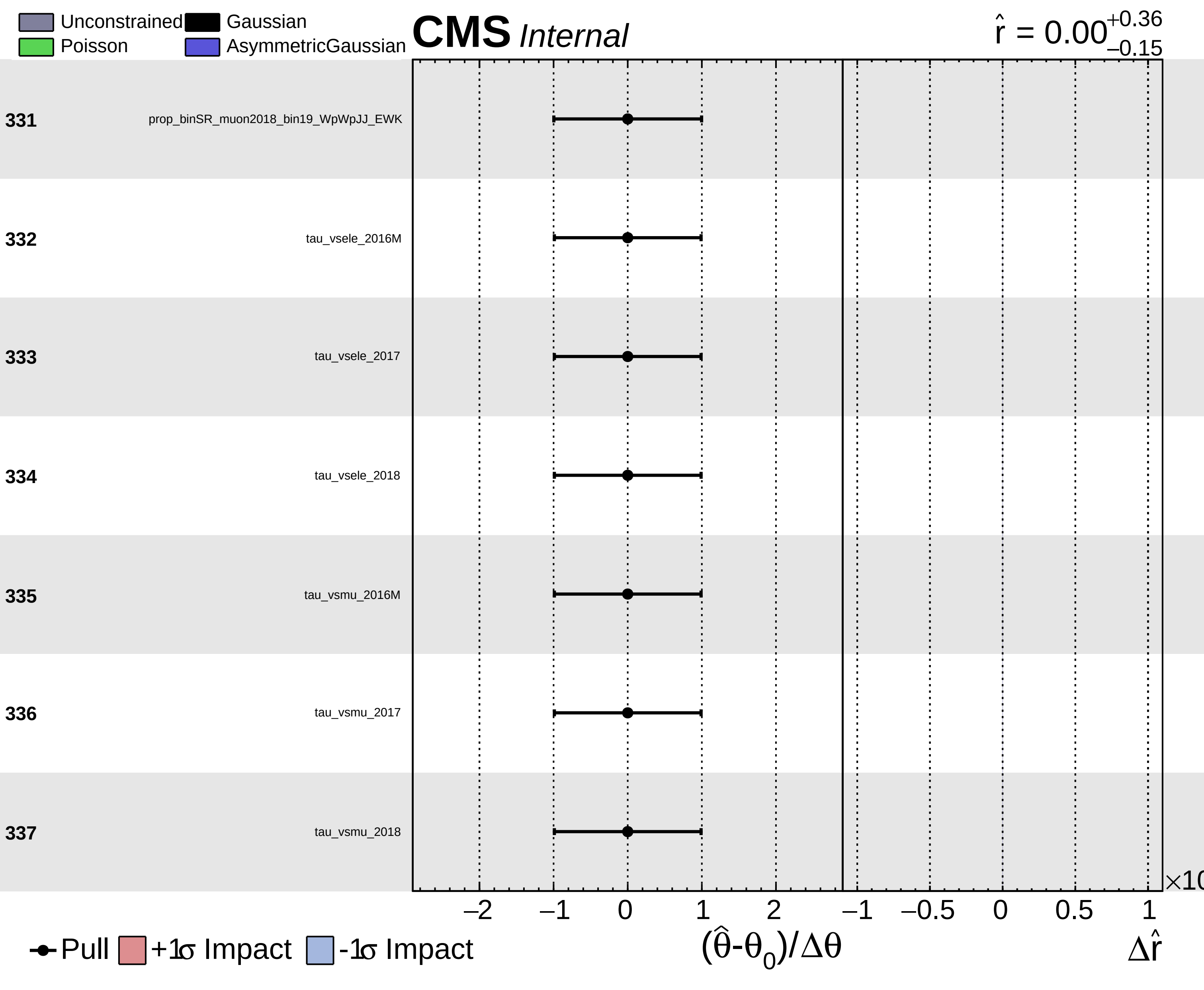


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
241 prop_binSR_electron2016M_bin19_WpWpJJ_QCD
242 prop_binSR_muon2016M_bin19_Triboson
243 prop_binSR_electron2017_bin21_WpWpJJ_QCD
244 lep_2018
245 prop_binSR_electron2016M_bin19_Triboson
246 prop_binSR_electron2016M_bin19_TVX
247 prop_binSR_electron2016M_bin20_TVX
248 lep_2016M
249 prop_binSR_electron2018_bin17_ZZtoLep
250 prop_binSR_electron2017_bin4
251 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
252 prop_binSR_electron2018_bin17_TVX
253 prop_binSR_muon2016M_bin20_Triboson
254 prop_binSR_electron2016M_bin19_WrongSign
255 prop_binSR_electron2017_bin21_TVX
256 prop_binSR_electron2016M_bin21_ZZtoLep
257 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
258 prop_binSR_electron2016M_bin15_TVX
259 prop_binSR_electron2016M_bin2_WZ
260 prop_binSR_electron2018_bin17_WpWpJJ_QCD
261 prop_binSR_electron2018_bin21_ZZtoLep
262 prop_binSR_electron2018_bin17_Triboson
263 prop_binSR_electron2016M_bin15_WrongSign
264 prop_binSR_electron2016M_bin13_Triboson
265 prop_binSR_electron2016M_bin15_WpWpJJ_QCD
266 prop_binSR_electron2016M_bin13_TVX
267 prop_binSR_electron2016M_bin15_Triboson
268 prop_binSR_electron2016M_bin21_WrongSign
269 prop_binSR_electron2017_bin16_WpWpJJ_QCD
270 prop_binSR_muon2016M_bin20_TVX
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
271 prop_binSR_electron2016M_bin2_TTTo2L2Nu
272 prop_binSR_electron2016M_bin3_TTTo2L2Nu
273 prop_binSR_electron2017_bin16_Triboson
274 prop_binSR_electron2016M_bin20_ZZtoLep
275 prop_binSR_electron2016M_bin4_WZ
276 prop_binSR_electron2016M_bin3_VG
277 prop_binSR_electron2016M_bin4_VG
278 prop_binSR_electron2016M_bin20_WrongSign
279 prop_binSR_electron2016M_bin4_TTTo2L2Nu
280 prop_binSR_muon2016M_bin19_ZZtoLep
281 prop_binSR_muon2016M_bin15_ZZtoLep
282 prop_binSR_muon2017_bin9_ZZtoLep
283 prop_binSR_electron2017_bin16_TVX
284 prop_binSR_electron2016M_bin2_ZZtoLep
285 prop_binSR_electron2017_bin21_ZZtoLep
286 prop_binSR_muon2017_bin11_TVX
287 prop_binSR_electron2016M_bin3_ZZtoLep
288 prop_binSR_muon2017_bin11_WpWpJJ_QCD
289 prop_binSR_electron2016M_bin13_ZZtoLep
290 prop_binSR_electron2016M_bin15_ZZtoLep
291 prop_binSR_electron2016M_bin2_TVX
292 prop_binSR_muon2017_bin11_Triboson
293 prop_binSR_electron2016M_bin4_ZZtoLep
294 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
295 prop_binSR_electron2016M_bin3_TVX
296 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
297 prop_binSR_electron2016M_bin2_Triboson
298 prop_binSR_electron2017_bin16_WrongSign
299 prop_binSR_electron2016M_bin4_WpWpJJ_QCD
300 prop_binSR_electron2016M_bin3_Triboson
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.15}$
301 prop_binSR_electron2016M_bin4_TVX
302 prop_binSR_electron2016M_bin4_Triboson
303 FES
304 QCDScale_WZ
305 QCDScale_sig
306 VBS_2016M
307 VBS_2017
308 VBS_2018
309 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
310 prop_binSR_electron2016M_bin15_WpWpJJ_EWK
311 prop_binSR_electron2016M_bin19_WpWpJJ_EWK
312 prop_binSR_electron2016M_bin20_WpWpJJ_EWK
313 prop_binSR_electron2016M_bin21_WpWpJJ_EWK
314 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
315 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
316 prop_binSR_electron2016M_bin4_WpWpJJ_EWK
317 prop_binSR_electron2017_bin16_WpWpJJ_EWK
318 prop_binSR_electron2017_bin19_WpWpJJ_EWK
319 prop_binSR_electron2017_bin21_WpWpJJ_EWK
320 prop_binSR_electron2018_bin17_WpWpJJ_EWK
321 prop_binSR_electron2018_bin21_WpWpJJ_EWK
322 prop_binSR_muon2016M_bin15_WpWpJJ_EWK
323 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
324 prop_binSR_muon2016M_bin19_WpWpJJ_EWK
325 prop_binSR_muon2016M_bin20_WpWpJJ_EWK
326 prop_binSR_muon2017_bin11_WpWpJJ_EWK
327 prop_binSR_muon2017_bin17_WpWpJJ_EWK
328 prop_binSR_muon2017_bin18_WpWpJJ_EWK
329 prop_binSR_muon2017_bin20_WpWpJJ_EWK
330 prop_binSR_muon2017_bin9_WpWpJJ_EWK
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.15
331
prop_binSR_muon2018_bin19_WpWpJJ_EWK
332
tau_vsele_2016M
333
tau_vsele_2017
334
tau_vsele_2018
335
tau_vsmu_2016M
336
tau_vsmu_2017
337
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂